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Roll No

CE-7101 (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Computational Methods in Structural Engineering

Time : Three Hours

Maximum Marks :70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Derive element stiffness matrix for a beam element in x - y plane.
2. Explain analysis procedure for pin jointed frame using direct stiffness method.
3. Explain the various steps of F.E.M.
4. Differentiate clearly between.
 - i) Matrix and conventional methods.
 - ii) Frames and grids.
 - iii) Stiffness and flexibility method.
5. Explain principal of contragradience and derive mass and stiffness matrices for an undamped three storey building having masses m_1 , m_2 , m_3 and stiffness k_1 , k_2 , k_3 respectively.
6. Derive the Jacobian matrix for a four noded isoperimetric quadrilateral element.
7. Explain with example how reduction of order of stiffness matrix is achieved?
8. Answer any two of the following:
 - a) Differentiate clearly between basic and direct stiffness method.
 - b) Explain elastic stability of 2-D rigid jointed frames.
 - c) State various penalty methods.
 - d) Explain convergence criteria for finite element analysis.
